

Student Number: _____ Class Teacher: _____

St George Girls High School

Trial Higher School Certificate Examination

2018



Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided at the back of this paper

Section II Pages 6 – 14

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section
- Begin each question in a new writing booklet

Section I	/10
Section II	
Question 11	/15
Question 12	/15
Question 13	/15
Question 14	/15
Question 15	/15
Question 16	/15
Total	/100

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I

10 marks

Attempt Questions 1 - 10

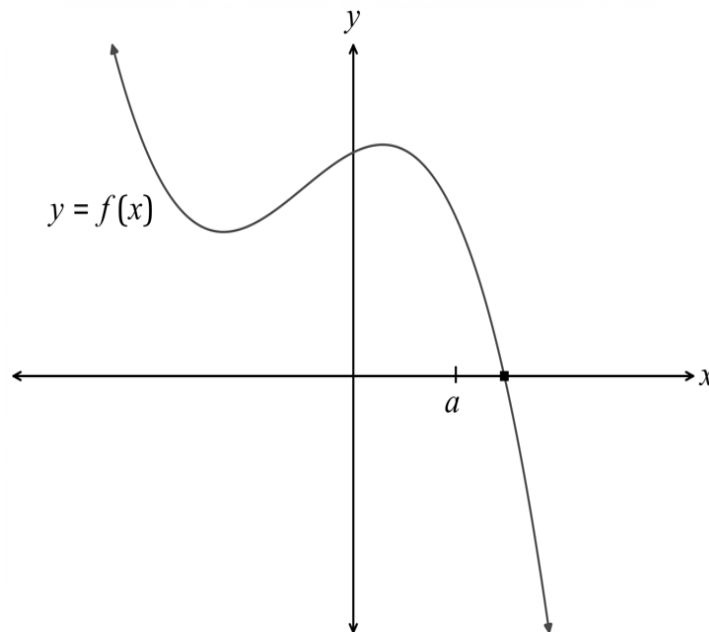
Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10.

1. The first three terms of an arithmetic sequence are 2, 6 and 10.
What is the 15th term of this sequence?

- (A) 58
- (B) 62
- (C) 450
- (D) 480

2. The diagram below shows the graph of $y = f(x)$.



Which of the following statements is true for $x = a$?

- (A) $f'(a) < 0$ and $f''(a) > 0$
- (B) $f'(a) > 0$ and $f''(a) > 0$
- (C) $f'(a) < 0$ and $f''(a) < 0$
- (D) $f'(a) > 0$ and $f''(a) < 0$

Section I (cont'd)

3. What is the area enclosed between $y = -\sqrt{1-x^2}$ and the x -axis from $x = -1$ to $x = 1$?

(A) $\frac{1}{2}\pi$

(B) π

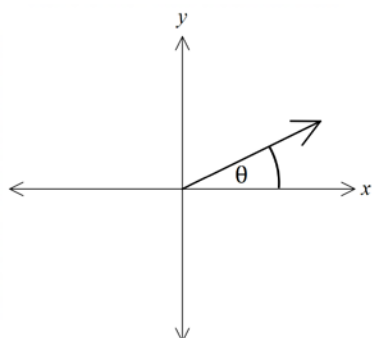
(C) 2π

(D) 4π

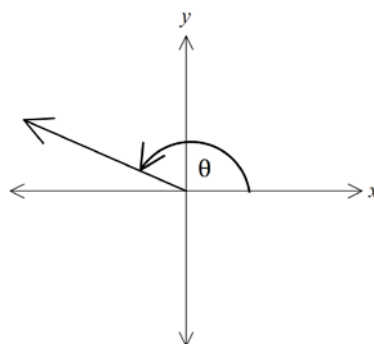
4. For the angle θ , $\sin \theta = -\frac{8}{17}$ and $\tan \theta = -\frac{8}{15}$.

Which diagram best shows angle θ ?

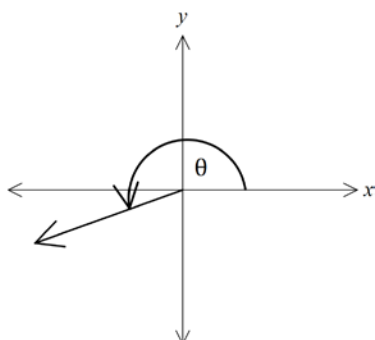
(A)



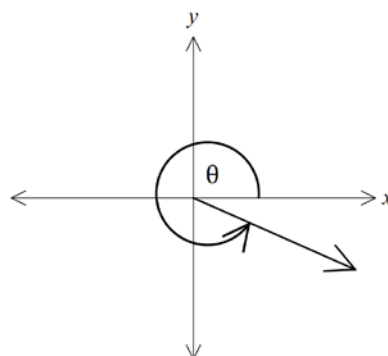
(B)



(C)



(D)



5. Find the limiting sum of the series $5 + \frac{5}{7} + \frac{5}{49} + \dots$

(A) ∞

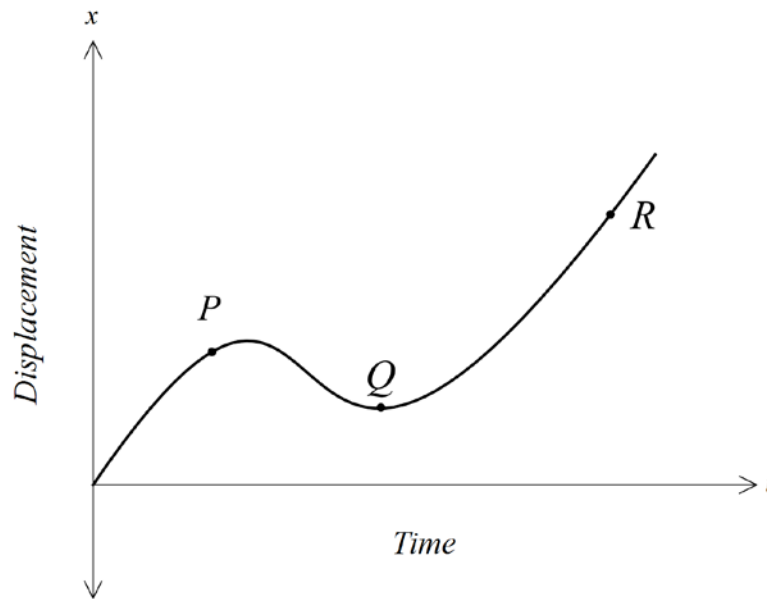
(B) $\frac{5}{42}$

(C) $5\frac{5}{6}$

(D) 6

Section I (cont'd)

6. The graph shows the displacement x of a particle moving along a straight line as a function of time t .



Which statement correctly describes the motion of the particle?

- (A) At point P , its acceleration and velocity are both positive.
- (B) At point P , its acceleration is negative while its velocity is positive.
- (C) At point Q , the particle is stationary and its acceleration is zero.
- (D) At point R , the particle is stationary and its acceleration is zero.
7. Find $\int \frac{1}{4x+1} dx$.
- (A) $\frac{-4}{(4x+1)^2} + C$
- (B) $4 \ln(4x+1) + C$
- (C) $\frac{1}{4} \ln(4x+1) + C$
- (D) $\ln(4x+1) + C$

Section I (Continued).

8. What is the derivative of $x + \ln x$?

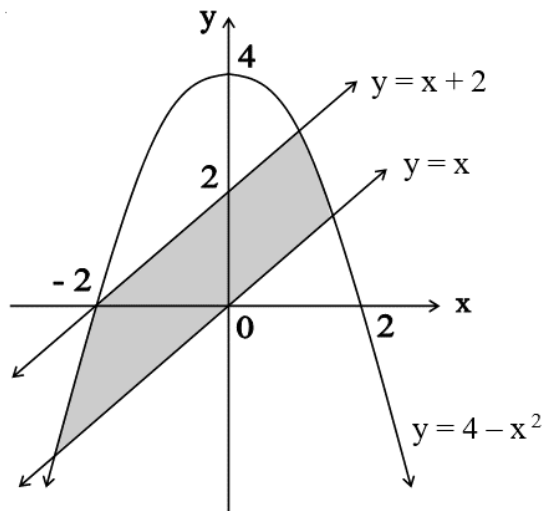
- (A) $1 + \frac{1}{x}$
- (B) $1 + \ln x$
- (C) $\frac{x^2}{2} + \frac{1}{x}$
- (D) $1 + \frac{1}{\ln x}$

9. Given that $\int_a^b f(x) dx = k$ and $\int_b^a g(x) dx = k - 2$.

What is the value of $\int_a^b [f(x) + g(x)] dx$?

- (A) 2
- (B) -2
- (C) $2k - 2$
- (D) $2 - 2k$

10. The shaded region in the diagram is enclosed by $y = 4 - x^2$, $y = x$ and $y = x + 2$.



Which of the following is the set of inequations that satisfy the shaded region?

- (A) $y \geq 4 - x^2, y \leq x, y \leq x + 2$
- (B) $y \leq 4 - x^2, y \leq x, y \leq x + 2$
- (C) $y \leq 4 - x^2, y \leq x, y \geq x + 2$
- (D) $y \leq 4 - x^2, y \geq x, y \leq x + 2$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section.

Start each question in a new writing booklet.

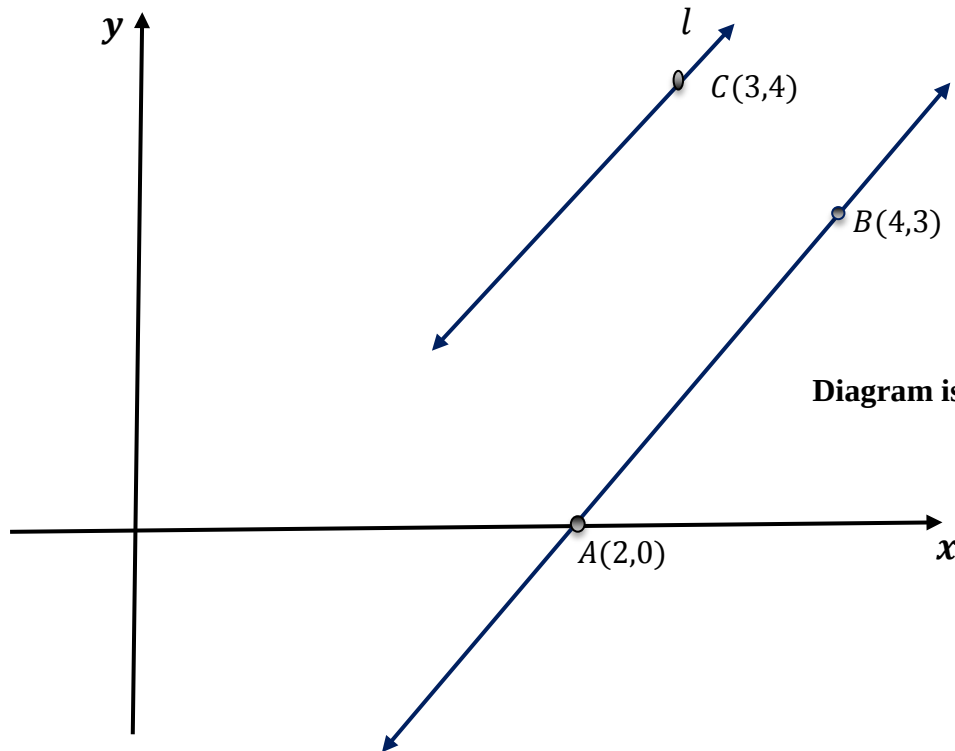
Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a New Writing Booklet.	Marks
a) Factorise $a^2 + a - 12$	1
b) Find the value of r given that $\frac{\sqrt{5}}{\sqrt{5} - 2} = 5 + r\sqrt{5}$.	2
c) Find $\int (\sin x + \cos x) dx$.	1
d) Differentiate $y = x \tan x$.	2
e) Solve $ x - 5 > 2$.	2
f) Find the coordinates of the focus of the parabola $(x + 3)^2 = 12(y - 1)$.	2
g) Find the domain of the function $f(x) = \ln(9 - x)$.	1
h) Find the equation of the tangent to the curve $y = x^3 - 2x$ at the point $(1, -1)$.	2
i) Find $\int_0^1 (1 + e^{-x}) dx$.	2

Question 12 (15 marks) Start a New Writing Booklet.

Marks

- a) In the diagram below, the points $A(2,0)$, $B(4,3)$ and $C(3,4)$ are shown.

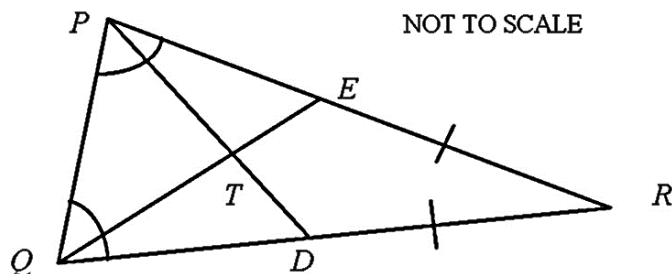


- | | | |
|-------|--|---|
| (i) | What is the exact length of AB ? | 1 |
| (ii) | Show that the equation of AB is $3x - 2y - 6 = 0$. | 1 |
| (iii) | Find the exact perpendicular distance from C to AB . | 1 |
| (iv) | The line l passing through C has the equation $3x - 2y - 1 = 0$.
(DO NOT PROVE THIS).
Show that the line l is parallel to AB . | 2 |
| (v) | D is a point on the line l such that the length DC is $\frac{\sqrt{13}}{2}$ units.
What type of quadrilateral is $ABCD$?
Give a reason for your answer. | 1 |
| (vi) | Calculate the area of $ABCD$. | 1 |

Question 12 (Continued).

Marks

- b) In the diagram $ER = DR$ and $\angle QPR = \angle PQR$. QE and PD intersect at T .



- (i) Copy this diagram into your writing booklet.

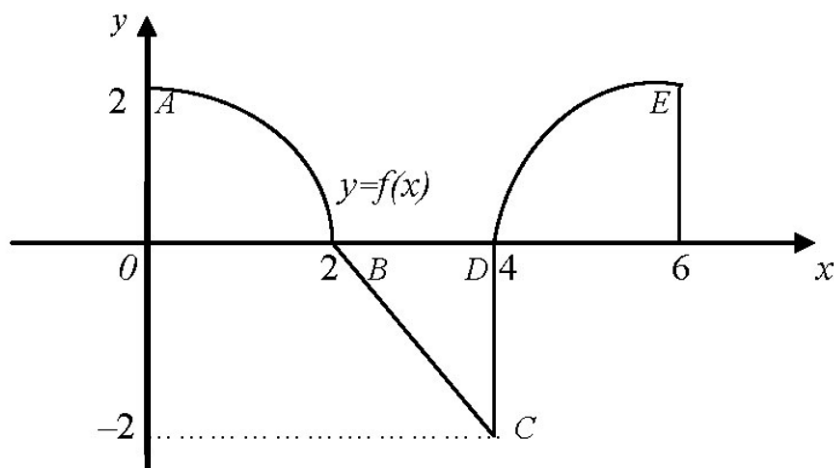
Prove, with full reasoning, that $\triangle QEP$ is congruent to $\triangle PDQ$.

3

- (ii) Why is $\triangle QTP$ isosceles?

2

- c) The graph of $y = f(x)$ consists of a quarter circle AB , triangle BCD and quarter circle DE as shown in the diagram below.



- (i) Evaluate $\int_0^6 f(x) dx$.

2

- (ii) State the values of x satisfying $0 < x < 6$ to indicate where

$y = f(x)$ is *not* differentiable.

1

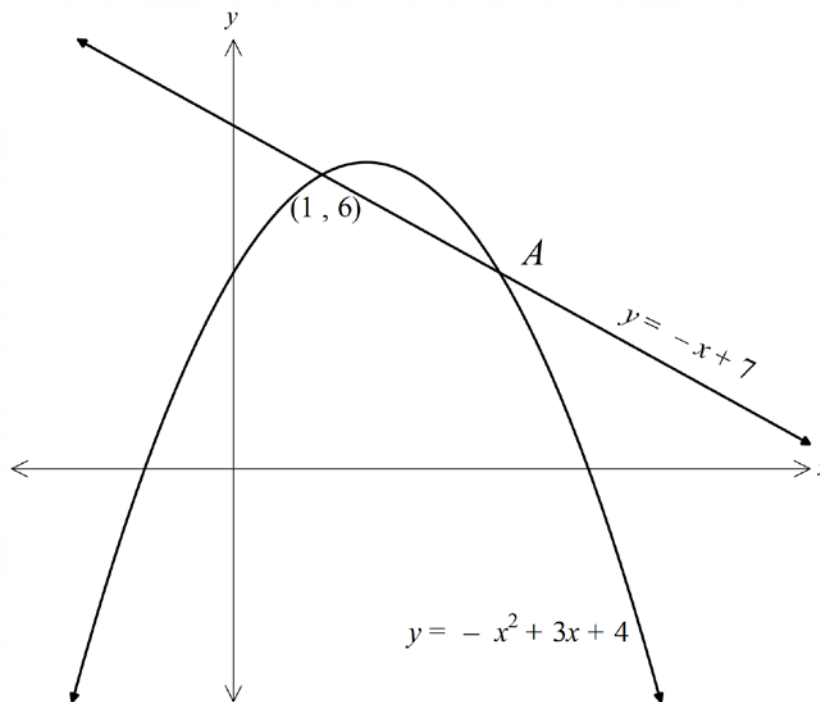
Question 13 (15 marks) Start a New Writing Booklet.

Marks

- a) Draw a neat sketch of $y = 3 \sin 4x$ for $0 \leq x \leq \pi$.
 Show clearly all of the relevant features.

2

- b) The parabola $y = -x^2 + 3x + 4$ and the line $y = -x + 7$ intersect at the point $(1, 6)$ and at point A .



- (i) Show that the x -coordinate of point A is 3. 2
- (ii) Calculate the area enclosed by the parabola and the line. 3
- c) Consider the function $f(x) = x^3 - 3x^2 - 24x + 20$.
- i) Find the coordinates of the turning points of the curve and determine their nature. 3
- ii) Find the values of x for which the curve is decreasing. 1
- iii) Sketch the curve showing the turning points and the y -intercept. 2
- d) Solve the equation $2(\ln x)^2 - \ln x - 1 = 0$.
 Leave your answer in exact form. 2

Question 14 (15 marks) Start a New Writing Booklet.

Marks

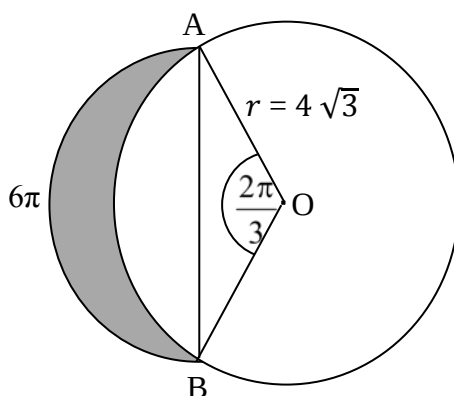
a) Solve $2 \cos 2\theta = 1$, for $0 \leq \theta \leq 2\pi$.

2

b) In the diagram, AB is a chord of a circle with centre O and radius $r = 4\sqrt{3}$, such that $\angle AOB = \frac{2\pi}{3}$.

AB is also the diameter of a semicircle with arc length 6π cm.

The length of $AB = 12$ cm.



Find the shaded area, **in exact form**, which lies inside the semicircle, but outside the circle.

3

c) At a particular location, a river 24 metres wide is measured for depth every 6 metres across its width. The measurements from bank to bank are given in the following table:

Distance across the river (m)	0	6	12	18	24
Depth (m)	0	8	22	6	0

(i) Use Simpson's rule to find the cross-sectional area of the river at this point.

2

(ii) Use your answer in part (i) to find the volume of water passing through this point in 3 hours, if the water passing this point travels at $\frac{1}{4} \text{ m/sec}$.

1

d) If α and β are the roots of $5x^2 - 2x + 6 = 0$, find the values of:

(i) $\alpha + \beta$.

1

(ii) $(\alpha + 1)(\beta + 1)$.

1

Question 14 (Continued).

Marks

- e) The initial size of a new bee colony was registered at 120000.
The number of bees B , in the colony was represented by $B = B_0 e^{0.5t}$, where t is in hours.
- (i) How many more bees were added to the colony during the first 5 hours? 2
- (ii) How fast was the colony growing at the 5-hour mark? 2
- (iii) How long did it take for the new colony to double in size? 1

Question 15 (15 marks) Start a New Writing Booklet.

Marks

a) Evaluate $\lim_{x \rightarrow 2} \left(\frac{x^3 - 8}{x - 2} \right)$. 2

b) Find all values of k , if $(k - 2), \sqrt{3k}, (k + 2)$ form a geometric progression. 3

c) Evaluate $\int_0^{\frac{\pi}{4}} \frac{dx}{\cos^2 x}$. 2

d) A particle moves along the x -axis with velocity $v(t)$ metres per second given by $v(t) = 16t - t^3$ after time t seconds.
At time $t = 2$, the displacement of the particle was 15 metres to the right of the origin.

i) Write an expression for the acceleration $a(t)$ of the particle. 1

ii) Find an expression for the displacement $s(t)$ of the particle. 2

iii) Find the total distance travelled between the times $t = 2$ and $t = 6$. 1

e) i) Prove 2

$$\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta} = 2 \operatorname{cosec}^2 \theta.$$

ii) Hence or otherwise solve 2

$$\operatorname{cosec} \theta \left[\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta} \right] = 16, \text{ for } 0^\circ \leq \theta \leq 2\pi.$$

Question 16 (15 marks) Start a New Writing Booklet.

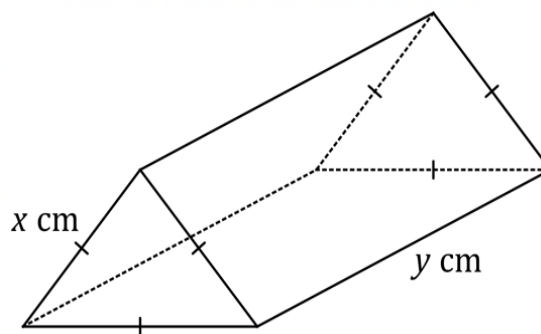
Marks

- a) A large tank of liquid which contains L litres of a toxic chemical is being drained.
 The amount of chemical in the tank over time t minutes, is given by:

$$L = 110(20 - t)^2.$$

- | | | |
|-----|---|---|
| i) | At what rate is the water draining out of the tank after 5 minutes? | 2 |
| ii) | How long will it take for the tank to be completely empty? | 1 |

- b) A triangular prism has a base that is an equilateral triangle with a side length of x cm.
 The length of the triangular prism is y cm.
 The volume of the prism is 1000 cm^3 .



Given the expression of y in terms of x is:

$$y = \frac{4000}{x^2\sqrt{3}} \quad (\text{DO NOT PROVE THIS})$$

- | | | |
|-----|---|---|
| i) | Show that the surface area, $A \text{ cm}^2$ of the prism is given by: | 2 |
| | $A = \frac{4000\sqrt{3}}{x} + \frac{\sqrt{3}x^2}{2}$ | |
| ii) | What is the value of x for the prism that will minimise the surface area? | 3 |

Question 16 (Continued).

Marks

- c) The region bounded by $y = p^2 - x^2$ and the x -axis, where p is a constant, is rotated about the y -axis between $y = 0$ to $y = p^2$ to form a solid.

Find the volume of this solid, in terms of p .

2

- d) Jane borrows \$60 000 to buy a new car.

The interest rate charged on the loan is 0.08% per week compounding weekly.

She agrees to repay the loan in equal fortnightly repayments of \$1000 each.

Let A_n be the amount of in dollars owing after her n^{th} fortnightly repayment.

- i) Show that $A_2 = \$60\,000 \times 1.008^4 - 1000(1 + 1.0008^2)$.

1

- ii) Using part (i), or otherwise, show that $A_n = \$624\,750.1 - \$564\,750.1 \times 1.0008^{2n}$.

2

- iii) How many weeks would it take Jane to repay her loan?

2

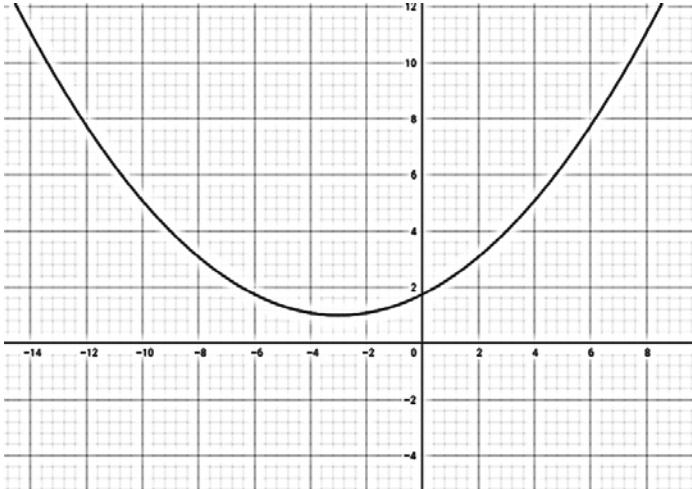
END OF THE EXAMINATION

Section 1

Question	Solution	Marking Guideline
1	$a = 2$ and $d = 4$ $T_n = a + (n - 1)d$ $T_{15} = 2 + (15 - 1) \times 4$ $= 58$	1 Mark: A
2	When $x = a$ the curve is decreasing and concave down. $\therefore f'(a) < 0$ and $f''(a) < 0$	1 Mark: C
3	The required area is a semicircle with a radius of 1 unit. $A = \frac{1}{2}\pi r^2 = \frac{1}{2} \times \pi \times 1^2$ $= \frac{1}{2}\pi$ square units	1 Mark: A
4	sin and tan are both negative in the fourth quadrant	1 Mark: D
5	$5 + \frac{5}{7} + \frac{5}{49} + \dots$ $\frac{a}{1-r} = \frac{5}{1-\frac{1}{7}} = 5\frac{5}{6}$	1 Mark: C
6	At point P, the slope of the curve is positive, therefore the velocity is positive. Concavity is negative, so acceleration is negative.	1 Mark: B
7	$\int \frac{1}{4x+1} dx$ $= \frac{1}{4} \int \frac{4}{4x+1} dx$ $= \frac{1}{4} \ln(4x+1) + C$	1 Mark: C
8	As $y = x + \ln x$, then $\frac{dy}{dx} = 1 + \frac{1}{x}$.	1 Mark: A
9	If $\int_b^a g(x) dx = k - 2$ then $\int_a^b g(x) dx = 2 - k$ Hence, $\int_a^b (f(x) + g(x)) dx = k + 2 - k = 2$	1 Mark: A
10	Select a point to test in the inequalities to decide which are correct. Test (0, 0) in $y \leq 4 - x^2$: $0 \leq 4 - 0$, true Test (0, 0) in $y \leq x + 2$: $0 \leq 0 + 2$, true Test (0, 1) in $y \leq x$: $1 \leq 0$, false, so $y \geq x$ Hence, option C contains the 3 correct inequalities.	1 Mark: D

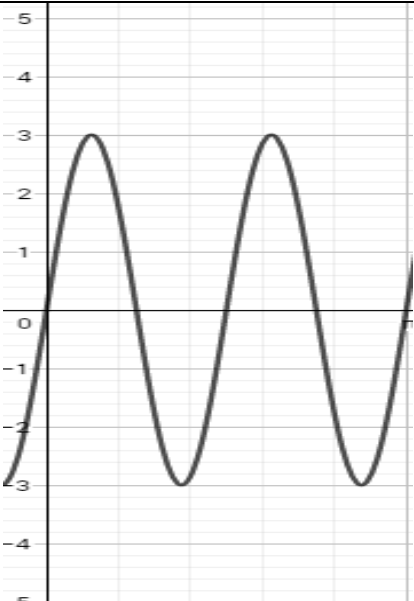
Section 2

Q11	Solution	Marking Guidelines
a	$a^2 + a - 12$ $= a^2 + 4a - 3a - 12$ $= a(a + 4) - 3(a + 4)$ $= (a - 3)(a + 4)$	1 Mark for correct factors.
b	$\frac{\sqrt{5}}{\sqrt{5} - 2} \times \frac{\sqrt{5} + 2}{\sqrt{5} + 2}$ $= \frac{5 + 2\sqrt{5}}{5 - 4}$ $= 5 + 2\sqrt{5}$ So $r = 2$	1 Mark for rationalising the denominator . 1 Mark for correct value of r .
c	$\int (\sin x + \cos x) dx$ $= -\cos x + \sin x + C$	1 Mark for correct integration.
d	Using the product rule: $Let u = x \text{ and } v = \tan x$ $u' = 1 \text{ and } v' = \sec^2 x$ $\frac{dy}{dx} = u'v + v'u$ $= \tan x \times 1 + x \sec^2 x$ $= \tan x + x \sec^2 x$	1 Mark for correct application of the product rule. 1 Mark for appropriate simplification.
e	$ x - 5 > 2$ $x - 5 > 2 \quad \text{or} \quad -x + 5 > 2$ $x > 7 \quad \quad \quad -x > -3$ $x > 7 \quad \quad \quad x < 3$	1 Mark for correct inequalities set up using concept of absolute value. 1 mark for correct solution.
f	The equation is in the form $(x - h)^2 = 4a(y - k)$ $(x + 3)^2 = 12(y - 1)$ Vertex = $(-3, 1)$ $4a = 12$ $a = 3$	1 Mark for vertex and focal length.

	 Hence, the co-ordinates of the focus are (-3,4)	1 Mark for correct coordinates of the focus.
g	$f(x) = \ln(9 - x)$ is only defined when $9 - x > 0$ $-x > -9$ $x < 9$ The domain is: $x < 9 \text{ or}$ $\text{all real } x : x < 9$	1 Mark for the correct domain.
h	$y = x^3 - 2x$ $\frac{dy}{dx} = 3x^2 - 2$ When $x = 1$, the tangent will have gradient $m = 3 \times 1 - 2$ $m = 1$ The equation of the tangent will be $y - y_1 = m(x - x_1)$ $y - (-1) = 1(x - 1)$ $y + 1 = x - 1$ $y = x - 2$	1 Mark for differentiating and finding the gradient. 1 Mark for finding the equation of the tangent.
h	$\int_0^1 (1 + e^{-x}) dx = [x - e^{-x}]_0^1$ $= (1 - e^{-1}) - (0 - e^0)$ $= 1 - \frac{1}{e} + 1$ $= 2 - \frac{1}{e}$	1 Mark for correct integration. 1 Mark for appropriate substitution and simplification.

Q12	Solution	Marking Guidelines
a (i)	$AB = \sqrt{(4-2)^2 + (3-0)^2}$ $= \sqrt{13}$	1 Mark for correct answer.
a (ii)	$\frac{y-0}{x-2} = \frac{3-0}{4-2}$ $2y = 3x - 6$ $3x - 2y - 6 = 0$	1 Mark for correct equation.
a (iii)	$d = \left \frac{3(3) - 2(4) - 6}{\sqrt{3^2 + 2^2}} \right $ $= \left \frac{-5}{\sqrt{13}} \right $ $= \frac{5}{\sqrt{13}} \text{ or } \frac{5\sqrt{13}}{13}$	1 Mark for correct perpendicular distance.
a (iv)	$AB: 2y = 3x - 6$ $y = \frac{3}{2}x - 3$ $m_1 = \frac{3}{2} \quad \text{or} \quad m_1 = -\frac{a}{b} = \frac{3}{2}$ Similarly, for l : $2y = 3x - 1$ $y = \frac{3}{2}x - \frac{1}{2}$ $m_2 = \frac{3}{2} \text{ or } m_2 = -\frac{a}{b} = \frac{3}{2}$ $\therefore AB // \text{line } l: (\text{gradients are equal})$	1 Mark for finding the gradient of AB or line M . 1 mark for correct justification.
a (v)	$ABCD$ is trapezium. (One pair of opposite sides are parallel, but not equal)	1 Mark for correct reasoning.
a(vi)	$A = \frac{1}{2}(a+b)h$ $= \frac{1}{2}\left(\frac{\sqrt{13}}{2} + \sqrt{13}\right) \times \frac{5}{\sqrt{13}}$ $= \frac{15}{4} \text{ or } 3.75 \text{ square units}$	1 Mark for correct area.
b(i)	In $\triangle QEP$ and $\triangle QDP$, $PR = QR$ (sides opposite equal angles) $ER = DR$ (given) $\therefore PE = QD$ (by subtraction of sides) QP is common $\angle QPE = \angle PQD$ (given) $\therefore \triangle QEP \equiv \triangle QDP$ (SAS holds)	1 Mark for justification of subtraction of sides. 1 Mark for the other 2 reasons. 1 Mark for appropriate congruency test.
b(ii)	$\triangle TEP \equiv \triangle QDT$, $\angle QTD = \angle PTE$ (vertically opposite angles are equal) $\angle QDT = \angle PET$ (corresponding angles in congruent triangles,	1 Mark for appropriate reasoning.

	$\triangle QEP \equiv \triangle QDP$ $QD = PE$ (already proven) Since $\triangle TEP \equiv \triangle TDQ$ (AAS) $PT = QT$ (corresponding sides of congruent triangles) $\therefore \triangle QTP$ is isosceles (2 equal sides) Or $\angle EQP = \angle DPQ$ (corresponding angles in congruent triangles) $\therefore \triangle TPQ$ is isosceles (angles opposite equal sides are equal)	1 Mark for appropriate justification of an isosceles triangle.
c (i)	$\int_0^6 f(x) \, dx = \frac{1}{4} \text{ circle} - \text{triangle} + \frac{1}{4} \text{ circle}$ $= \frac{1}{4} \pi (4) - \frac{1}{2} (2 \times 2) + \frac{1}{4} \pi (4)$ $= 2\pi - 2$	1 Mark for appropriate working. 1 Mark for correct value of the integral.
c (ii)	$x = 2$ and $x = 4$	1 Mark for the appropriate points.

Q13	Solution	Marking Guidelines
a	 Amplitude = 3 Period = $\frac{\pi}{2}$	1 Mark for correct shape. 1 Mark for correct amplitude and period. Some students were confused between the sine and cosine curve.
b (i)	$y = -x^2 + 3x + 4 \quad (1)$ $y = -x + 7 \quad (2)$ $-x^2 + 3x + 4 = -x + 7 \text{ (sub (1) into (2))}$ $-x^2 + 4x - 3 = 0$ $x^2 - 4x + 3 = 0$ $(x - 3)(x - 1) = 0$ $\therefore x = 1 \text{ (which we already knew) and } x = 3$ $\therefore \text{ point A has x-coordinate 3.}$	1 Mark for correct use of simultaneous equations. 1 Mark for correct x-coordinate. Most students failed to set up simultaneous equations appropriately.
b (ii)	$A = \int_1^3 (-x^2 + 3x + 4) - (-x + 7) \, dx$ $= \int_1^3 (-x^2 + 4x - 3) \, dx$ $= \left[-\frac{x^3}{3} + 2x^2 - 3x \right]_1^3$ $= \left[\left(-\frac{27}{3} + 2(3^2) - 3(3) \right) - \left(-\frac{1}{3} + 2(1^2) - 3(1) \right) \right]$ $= \left[0 + \frac{4}{3} \right]$ $= \frac{4}{3} \text{ square units.}$	1 Mark for setting up correct integral. 1 Mark for correct integration. 1 Mark for correct answer. The majority of students attempted this question very well.

c (i)	$f(x) = x^3 - 3x^2 - 24x + 20$ $f'(x) = 3x^2 - 6x - 24$ Let $f'(x) = 0$ to find the possible stationary turning points, we get: $3x^2 - 6x - 24 = 0$ that is $x^2 - 2x - 8 = 0$ so $(x - 4)(x + 2) = 0$ Hence, $x = 4$ or $x = -2$ When $x = 4$, $y = -60$ and when $x = -2$, $y = 48$ In table, <table><tr><td>x</td><td>-3</td><td>-2</td><td>0</td><td>4</td><td>5</td></tr><tr><td>$f'(x)$</td><td>21</td><td>0</td><td>-24</td><td>0</td><td>21</td></tr><tr><td>$f(x)$</td><td></td><td>(-2, 48) max</td><td></td><td>min (4, -60)</td><td></td></tr></table> $f'(-3) > 0$ and $f'(0) < 0$ So the curve has a maximum at $(-2, 48)$. Also, as $f'(0) < 0$ and $f'(5) > 0$ then the curve has a minimum at $(4, -60)$. Alternative method: $f''(x) = 6x - 6$ $f''(-2) = -18$ Hence max turning point at $(-2, 48)$. $f''(4) = 18$ Hence min turning point at $(4, -60)$.	x	-3	-2	0	4	5	$f'(x)$	21	0	-24	0	21	$f(x)$		(-2, 48) max		min (4, -60)		1 Mark for correct differentiation and making the expression equal to zero. 1 Mark for determining the stationary points. 1 Mark for determining the nature of the stationary points. The majority of students attempted this question very well.
x	-3	-2	0	4	5															
$f'(x)$	21	0	-24	0	21															
$f(x)$		(-2, 48) max		min (4, -60)																
c (ii)	The curve is decreasing when $f'(x) < 0$. Hence, it is decreasing for $-2 < x < 4$	1 Mark for correct values of x.																		
c (iii)		1 Mark for correct shape. 1 Mark for displaying the turning point and y-intercept. Well done.																		
d	$2\ln^2 x - \ln x - 1 = 0$ Let $k = \ln x$ we get $2k^2 - k - 1 = 0$ $(2k + 1)(k - 1) = 0$ So $k = -\frac{1}{2}$ or $k = 1$ Hence $\ln x = -\frac{1}{2}$ or $\ln x = 1$ $x = e^{-\frac{1}{2}} \left(ie \frac{1}{\sqrt{e}} \right) \text{ or } x = e$ Note both answers are valid as they verify the original equation.	1 Mark for correct values of k. 1 Mark for the correct values of x. Some students were confused with the log being negative compared to positive log.																		

Q14	Solution	Marking Guidelines
a	$2 \cos 2\theta = 1$ $\cos 2\theta = \frac{1}{2}$ $2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$ $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$	1 Mark for half the correct solutions. 1 Mark for the remaining solutions.
b	The shaded region = area of semicircle with diameter AB – area of minor segment with chord AB . $= \frac{1}{2} \times \pi \times 6^2 - \frac{1}{2} \times (4\sqrt{3})^2 \times (\frac{2\pi}{3} - \sin \frac{2\pi}{3})$ $= 18\pi - 24 \times (\frac{2\pi}{3} - \frac{\sqrt{3}}{2})$ $= 18\pi - 16\pi + 12\sqrt{3}$ $= (2\pi + 12\sqrt{3}) \text{ cm}^2$	2 Marks for appropriate calculation. 1 Mark for the correct answer.
c (i)	$A \approx \frac{h}{3} [y_0 + 2(y_2 + y_4 + \dots) + 4(y_1 + y_3 + \dots) + y_l]$ $A \approx \frac{6}{3} [0 + 2(22) + 4(8 + 6) + 0]$ $A \approx 200 \text{ m}^2$	1 Mark for correct application of Simpson's Rule. 1 Mark for correct area.
c (ii)	$\text{Volume} = 200 \text{ m}^2 \times \frac{1}{4} \text{ m/s} \times 3 \text{ hrs} \times 3600 \text{ s}$ $\text{Volume} = 540\,000 \text{ m}^3$	1 Mark for correct answer.
d (i)	$5x^2 - 2x + 6 = 0$ $\alpha + \beta = -\frac{b}{a} = \frac{2}{5}$	1 Mark for correct answer.
d (ii)	$(\alpha + 1)(\beta + 1)$ $= \alpha\beta + \alpha + \beta + 1$ $= \frac{6}{5} + \frac{2}{5} + 1 = \frac{13}{5}$	1 Mark for correct answer.
e (i)	$B = B_0 e^{0.5t}$ So $B = 120000 e^{0.5t}$ $B \text{ size} = 120000 e^{0.5(5)}$ At $t=5$ $B = 1461899$ approximately. Number added = $1461899 - 120000$ $= 1341899$ bees	1 Mark for correct substitution and attaining 1461899. 1 Mrk for the number added to the colony.
e (ii)	(b) (ii) $\frac{dB}{dt} = 60000 e^{0.5(5)} = 730949.6$	1 Mark for correct Differentiation 1 Mark for finding the correct rate.
	(iii) $240000 = 120000 e^{0.5t}$ $2 = e^{0.5t} \ln 2 = 0.5t$ $t = 2 \ln 2 = \ln 4 = 1.39 \text{ hrs}$	1 Mark for correct substitution and evaluation of time.

Q15	Solution	Marking Guidelines
a	$\lim_{x \rightarrow 2} \left(\frac{x^3 - 8}{x - 2} \right)$ $= \lim_{x \rightarrow 2} \left(\frac{(x - 2)(x^2 + 2x + 4)}{x - 2} \right)$ $= \lim_{x \rightarrow 2} (x^2 + 2x + 4)$ $= (2)^2 + 2(2) + 4$ $= 12$	1 Mark for factorising difference of cubes. 1 Mark for correct limit.
b	$\frac{k+2}{\sqrt{3k}} = \frac{\sqrt{3k}}{k-2}$ $k^2 - 4 = 3k$ $k^2 - 3k - 4 = 0$ $(k - 4)(k + 1) = 0$ $k = -1, 4$ $k > 0$ $\therefore k = 4$	1 Mark for using the correct test with correct substitution. 1 Mark for achieving $k = -1, 4$. 1 Mark for correct solution.
c	$\int_0^{\frac{\pi}{4}} \frac{dx}{\cos^2 x} = \int_0^{\frac{\pi}{4}} \sec^2 x \, dx$ $= [\tan x]_0^{\frac{\pi}{4}}$ $= 1 - 0$ $= 1$	1 Mark for achieving $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$. 1 Mark for correct solution.
d (i)	$\frac{dv}{dt} = a = 16 - 3t^2$	1 Mark for achieving $a = 16 - 3t^2$
d (ii)	$s(t) = \int (16t - t^3) \, dt$ $s = 15, t = 2$ $= 8t^2 - \frac{1}{4}t^4 + c$ $15 = 32 - 4 + c$ $\therefore c = -13$ $\text{So } s(t) = 8t^2 - \frac{1}{4}t^4 - 13$	1 Mark for achieving $= 8t^2 - \frac{1}{4}t^4 + c$ 1 Mark for achieving $s(t)$ $= 8t^2 - \frac{1}{4}t^4 - 13$
d (iv)	from $t=2$ to $t=4$ travels 36 metres $t=6$ travels -100 $\therefore$ total distance = 136	1 Mark for appropriate Working and answer.

e(i)	$\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta}$ $= \frac{1 + \cos \theta + 1 - \cos \theta}{(1 - \cos \theta)(1 + \cos \theta)}$ $= \frac{2}{(1 - \cos \theta)(1 + \cos \theta)}$ $= \frac{2}{1 - \cos^2 \theta}$ $= \frac{2}{\sin^2 \theta}$ $= 2 \operatorname{cosec}^2 \theta$	1 Mark for obtaining $\frac{2}{(1 - \cos \theta)(1 + \cos \theta)}$ 1 Mark for further simplification
e(ii)	$\operatorname{cosec} \theta \left[\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta} \right] = 16$ $\operatorname{cosec} \theta (2 \operatorname{cosec}^2 \theta) = 16$ $\operatorname{cosec}^3 \theta = 8$ $\operatorname{cosec} \theta = 2$ $\sin \theta = \frac{1}{2}$ $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$	1 Mark for obtaining $\operatorname{cosec}^3 \theta = 8$ 1 Mark for correct values of θ.

Q16	Solution	Marking Guidelines
a(i)	$L = 110(20 - t)^2$ $\therefore \text{Rate} = \frac{dL}{dt} = 110 \times 2(20 - t) \times -1$ $= -220(20 - t)$ $\therefore \text{at } t = 5$ $\frac{dL}{dt} = -220(20 - 5) = -3300 \text{ L / Min}$	1 Mark for correct differentiation. ½ mk for 3300 ½ mk for the units Ignored the sign as the tank was draining.
a(ii)	For empty tank : $L = 0$ i.e. $110(20 - t)^2 = 0$ $(20 - t)^2 = 0$ $\therefore t = 20 \text{ minutes}$ Or use $\frac{dL}{dt} = 0$	1 Mark for $L = 0$ and $t = 20$ minutes.
b (i)	$A = 3xy + 2 \left(\frac{1}{2} \times x \times x \times \sin 60^\circ \right)$ $= 3xy + 2 \left(\frac{\sqrt{3}}{4} x^2 \right)$ $= 3x \times \frac{4000}{x^2 \sqrt{3}} + 2 \left(\frac{\sqrt{3}}{4} x^2 \right)$ $= \frac{4000\sqrt{3}}{x} + \frac{\sqrt{3}x^2}{2}$ Or $V = 1000$ $1000 = \text{Area of triangle} \times y$ $\text{Area of triangle} = \frac{1000}{y}$ $\text{Surface Area} = \frac{2000}{y} + 3xy, \text{ etc}$	1 Mark for finding the correct expression for the surface area. Or use Pythagoras' Theorem 1 Mark for the correct simplification.
b (ii)	Minimal A occurs when $\frac{dA}{dx} = 0$ $A = \frac{4000\sqrt{3}}{x} + \frac{\sqrt{3}x^2}{2}$ $\frac{dA}{dx} = -4000\sqrt{3} \times x^{-2} + \sqrt{3}x$ Hence $-4000\sqrt{3} \times x^{-2} + \sqrt{3}x = 0$ $\frac{4000\sqrt{3}}{x^2} = \sqrt{3}x$	1 Mark for finding the derivative.

	$x = \sqrt[3]{4000}$ Check if a minima $\frac{d^2 A}{dx^2} = 8000\sqrt{3} \times x^{-3} + \sqrt{3}$ When $x = \sqrt[3]{4000}$ $\frac{d^2 A}{dx^2} = 8000\sqrt{3} \times (\sqrt[3]{4000})^{-3} + \sqrt{3}$ $= 2\sqrt{3} + \sqrt{3} > 0, \text{ therefore minimum.}$	1 Mark for finding the value of x without testing for a minima. 1/2 Mark for justifying whether the solution is maxima or minima. ½ mark for the answer (including 5.196...)
c	Volume of rotation about the y axis is: $V = \pi \int_a^b x^2 dy$ $V = \pi \int_0^{p^2} (p^2 - y) dy$ $= \pi \left[p^2 y - \frac{y^2}{2} \right]_0^{p^2}$ $= \pi \left[(p^4 - \frac{p^4}{2}) - (0 - 0) \right] = \frac{\pi p^4}{2} \text{ units}^3$	1 Mark for correct application of the volume formula. 1 Mark for correct answer. Note the $\int p^2 dy = p^2 y$
d (i)	$A_1 = \$60000 \times 1.0008^2 - 1000$ $A_2 = (\$60000 \times 1.0008^2 - 1000) \times 1.0008^2 - 1000$ $= \$60000 \times 1.0008^4 - 1000 \times 1.0008^2 - 1000$ $= \$60000 \times 1.0008^4 - 1000 \times (1 + 1.0008^2)$	1 Mark for showing the necessary steps.
d (ii)	$A_2 = \$60000 \times 1.0008^4 - 1000 \times (1 + 1.0008^2)$ $A_n = 60000 \times 1.0008^{2n} - 1000(1 + 1.0008^2 + \dots + 1.0008^{2n-2})$ $= 60000 \times 1.0008^{2n} - 1000 \left(\frac{(1.0008^2)^n - 1}{1.0008^2 - 1} \right)$ $= 60000 \times 1.0008^{2n} - 624750.1(1.0008^{2n} - 1)$ $= 60000 \times 1.0008^{2n} - 624750.1 \times 1.0008^{2n} + 624750.1$ $= 624750.1 - 564750.1 \times 1.0008^{2n}$	½ mark ½ mark ½ mark ½ for no errors leading to the answer. Note Many students did not use (1.0008^2) As the common ratio.

d (iii)	The loan will be repaid when $A_n = 0$ Solve: $0 = \$624750.1 - 564750.1 \times 1.0008^{2n}$ $564750.1 \times 1.0008^{2n} = 624750.1$ $1.0008^{2n} = 624750.1 \div 564750.1$ $1.0008^{2n} = 1.10624....$ $\ln(1.0008^{2n}) = \ln(1.10624....)$ $2n \times \ln(1.0008) = \ln(1.10624....)$ $2n = \frac{\ln(1.10624....)}{\ln(1.0008)}$ $n = \frac{\ln(1.10624....)}{2\ln(1.0008)}$ $n = 63.129 \text{ fortnights}$ $n = 126.26 \text{ weeks}$	1 Mark for appropriate substitution into the formula and simplification. 1/2 Mark for correct answer. ½ mark for correct time unit.
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